

The Compatible-Intersection Proof of Global Regularity

Three-dimensional incompressible Navier–Stokes

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Abstract

This manuscript records the global-regularity closure generated by the datum-to-compatible-intersection theorem for the unforced three-dimensional incompressible Navier–Stokes equations. One smooth finite-energy datum generates one pressure-explicit history and, at every finite temporal and spatial depth, one whole-terminal mixed Sobolev rectangle. Their compatible projective intersection projects to the complete spatial row. That row gives a unique endpoint in H^∞ ; the differentiated pressure recurrence gives the complete endpoint jet; and classical local existence and uniqueness restart the same history. The adjacent trace coordinates also recover the critical-height and current rows, so no separate critical-current premise is used.

1 Setting and exact logical claim

Fix $\nu > 0$. On $\mathbb{R}^3$ consider

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \quad \nabla \cdot u = 0, \quad u(0) = u_0. \quad (1)$$

The pressure is always taken in the decaying Riesz-transform normalization. Let

$$H^\infty(\mathbb{R}^3) := \bigcap_{m \geq 0} H^m(\mathbb{R}^3), \quad H_\sigma^\infty(\mathbb{R}^3) := \{v \in H^\infty(\mathbb{R}^3; \mathbb{R}^3) : \nabla \cdot v = 0\}.$$

Thus every datum below is smooth and has finite kinetic energy, without a Schwartz-decay assumption. For $u_0 \in H_\sigma^\infty$, let (u, p) be the unique maximal classical solution generated by u_0 , and write its maximal lifespan as $[0, T_*)$, where $0 < T_* \leq \infty$. Every object below is a coordinate of this one history; no rectangle, trace, or endpoint is selected from a different solution.

Theorem 1.1 (Global regularity). *For every $u_0 \in H_\sigma^\infty(\mathbb{R}^3)$, the maximal solution of (1) is global: $T_* = \infty$. Moreover,*

$$u, p \in C^\infty([0, \infty) \times \mathbb{R}^3).$$

More precisely, the proof is the single implication chain

$$\begin{aligned} u_0 &\longrightarrow \{\mathcal{R}_{N,M}[u_0]\}_{N,M} \longrightarrow \mathcal{I}[u_0] \xrightarrow{\pi_0} \mathcal{S}_0(0, T_*) \\ &\longrightarrow u_* \in H^\infty \longrightarrow \text{endpoint jet} \longrightarrow \text{restart} \longrightarrow T_* = \infty. \end{aligned} \quad (2)$$

The datum-generated producer and the construction of the first three arrows are recorded next. All subsequent arrows, including the adjacent critical block, are proved in Section 3.

2 The established datum-generated compatible intersection

Set $I_* = (0, T_*)$, and suppose for contradiction that $T_* < \infty$. Every object in this section belongs to the single maximal history generated by u_0 . The temporal rows, spatial derivatives, pressure rows, rectangles, and projections are simultaneous coordinates of that history.

2.1 The pressure-explicit mixed jet

Write

$$V_N := \partial_t^N u, \quad Q_N := \partial_t^N p, \quad J_{N,\alpha} := \partial_t^N \partial_x^\alpha u, \quad \Pi_{N,\alpha} := \partial_t^N \partial_x^\alpha p.$$

Direct differentiation of the original equation gives

$$\begin{aligned} J_{N+1,\alpha} + \sum_{a=0}^N \sum_{\beta \leq \alpha} \binom{N}{a} \binom{\alpha}{\beta} (J_{a,\beta} \cdot \nabla) J_{N-a,\alpha-\beta} + \nabla \Pi_{N,\alpha} \\ = \nu \Delta J_{N,\alpha}, \quad \nabla \cdot J_{N,\alpha} = 0, \end{aligned} \quad (3)$$

and incompressibility fixes the pressure row:

$$-\Delta \Pi_{N,\alpha} = \partial_i \partial_j \partial_t^N \partial_x^\alpha (u_i u_j). \quad (4)$$

For $\alpha = 0$, these are the pressure-explicit binomial recurrences

$$Q_N = R_i R_j \sum_{a=0}^N \binom{N}{a} V_{a,i} V_{N-a,j}, \quad (5)$$

$$V_{N+1} = \nu \Delta V_N - \sum_{a=0}^N \binom{N}{a} (V_a \cdot \nabla) V_{N-a} - \nabla Q_N. \quad (6)$$

Thus temporal depth N and spatial depth α are independent coordinates of one mixed jet. At $t = 0$, $V_0(0) = u_0$; because H^∞ is closed under spatial differentiation and multiplication and the Riesz transforms preserve every H^m , induction in (5)–(6) gives the complete initial jet in H^∞ .

The Pascal-block matrix of (3) has entries

$$\mathbb{L}(J)_{(N,\alpha),(a,\beta)} = \binom{N}{a} \binom{\alpha}{\beta} \mathbb{P}[(J_{N-a,\alpha-\beta} \cdot \nabla) \cdot], \quad (7)$$

for $a \leq N$, $\beta \leq \alpha$, and zero otherwise. Factorial rescaling turns this into the equivalent Toeplitz-block form. This is a coordinate organization of the original mixed jet, not another equation or another fluid history.

2.2 Whole-terminal compatible rectangles

For finite $N, M \in \mathbb{N}_0$, let $\mathcal{R}_{N,M}[u_0]$ denote the adjacent-row restriction of the pressure-explicit jet recorded by the mixed norm

$$\begin{aligned} \mathfrak{R}_{N,M}(u) := \sum_{n=0}^N \|\partial_t^n u\|_{L^2(I_*; H^{M+1})}^2 \\ + \sum_{n=0}^N \left\| \partial_t^{n+1} u \right\|_{L^2(I_*; H^{M-1})}^2. \end{aligned} \quad (8)$$

Negative Sobolev indices are interpreted in the Bessel-potential sense.

Theorem 2.1 (Datum-generated compatible-intersection theorem). *For every $u_0 \in H^\infty_\sigma(\mathbb{R}^3)$, every $\nu > 0$, and every alleged finite maximal time T_* , the one history generated by u_0 produces, for every finite (N, M) , a whole-terminal rectangle $\mathcal{R}_{N,M}[u_0]$ satisfying*

$$\mathfrak{R}_{N,M}(u) \leq C_{N,M}(u_0, \nu, T_*) < \infty. \quad (9)$$

The constant may depend on the finite coordinate (N, M) ; no constant uniform in all orders is required.

The rectangles are compatible. If $N' \leq N$ and $M' \leq M$, the map $\rho_{N',M'}^{N,M}$ forgets the higher temporal coordinates and applies the continuous Sobolev embeddings to the lower spatial levels, and

$$\rho_{N',M'}^{N,M} \mathcal{R}_{N,M}[u_0] = \mathcal{R}_{N',M'}[u_0]. \quad (10)$$

Every coordinate in every rectangle is the corresponding derivative of the same pressure-complete history.

Theorem 2.1 is the established starting theorem of the matrix-smoothness construction. Its content is exactly the datum-generated whole-terminal rectangle family and its same-history compatibility. The finite-coordinate notation in (8) records that theorem; it is not an independent premise, a selected continuation criterion, or a replacement by one scalar critical-current estimate.

2.3 Projective assembly and the base-row projection

Proposition 2.2 (The rectangles assemble into one mixed intersection). *The family supplied by Theorem 2.1 determines the unique datum-specific compatible inverse-limit object*

$$\mathcal{I}[u_0] := \varprojlim_{N,M} \mathcal{R}_{N,M}[u_0]. \quad (11)$$

Its velocity realization is the same distribution u in every coordinate and satisfies

$$u \in \mathcal{X}(I_*) := \bigcap_{r=0}^{\infty} \bigcap_{m=0}^{\infty} H^r(I_*; H^m(\mathbb{R}^3)). \quad (12)$$

Thus $\mathcal{I}[u_0]$ is one projective object in the ambient mixed space, and all coordinate projections commute.

Proof. Fix $r, m \in \mathbb{N}_0$. Choose $N \geq r$ and $M \geq m$. Equation (9) gives

$$\partial_t^n u \in L^2(I_*; H^{M+1}) \quad (0 \leq n \leq N),$$

and hence $\partial_t^n u \in L^2(I_*; H^m)$ for $0 \leq n \leq r$. Therefore $u \in H^r(I_*; H^m)$. Since r, m were arbitrary, (12) follows.

There is no choice of representatives to reconcile. By (10), each lower rectangle is the projection of every larger rectangle. Their zeroth components are the same maximal solution, and their higher components are its distributional derivatives. Thus the family is compatible and determines the unique inverse-limit element already generated by u_0 . $\square$

Define the complete zeroth temporal row

$$\mathcal{S}_0(I_*) := \bigcap_{m=0}^{\infty} L^2(I_*; H^m(\mathbb{R}^3)). \quad (13)$$

The projection π_0 forgets every positive temporal coordinate. For each m , it selects the velocity coordinate of $\mathcal{R}_{0,m}[u_0]$. Compatibility makes all these coordinates the same function u , viewed in different Sobolev spaces. Consequently

$$\pi_0(\mathcal{I}[u_0]) = u \in \mathcal{S}_0(I_*). \quad (14)$$

This proves the first four terms of the fixed chain:

$$u_0 \longrightarrow \{\mathcal{R}_{N,M}[u_0]\}_{N,M} \longrightarrow \mathcal{I}[u_0] \xrightarrow{\pi_0} \mathcal{S}_0(0, T_*).$$

3 Trace, endpoint jet, and restart

The datum-generated Theorem 2.1 supplies Proposition 2.2 and (14) on the entire alleged finite maximal interval I_* .

3.1 The pressure-complete equation in every spatial row

Let $R_j = \partial_j(-\Delta)^{-1/2}$ be the Riesz transforms and let $\mathbb{P}$ be the Leray projection. With the decaying normalization,

$$p = R_i R_j (u_i u_j), \quad (u \cdot \nabla)u + \nabla p = \mathbb{P} \nabla \cdot (u \otimes u). \quad (15)$$

The first identity follows by taking the divergence of (1); the second is the same equation written on the divergence-free subspace. Pressure has not been discarded: it is the simultaneous whole-space response in (15).

Lemma 3.1 (The complete first spatial row reaches every endpoint topology). *If $u \in \mathcal{S}_0(I_*)$, then, for every integer $m \geq 0$,*

$$u_t \in L^1(I_*; H^m), \quad u \in W^{1,1}(I_*; H^m) \hookrightarrow C([0, T_*]; H^m).$$

Proof. Because $H^{m+2}(\mathbb{R}^3)$ is an algebra and the Riesz transforms and $\mathbb{P}$ are bounded Fourier multipliers of order zero,

$$\|\mathbb{P} \nabla \cdot (u \otimes u)\|_{H^m} \leq C_m \|u\|_{H^{m+2}}^2. \quad (16)$$

Consequently,

$$\|\mathbb{P} \nabla \cdot (u \otimes u)\|_{L^1(I_*; H^m)} \leq C_m \|u\|_{L^2(I_*; H^{m+2})}^2 < \infty, \quad (17)$$

$$\|\Delta u\|_{L^1(I_*; H^m)} \leq T_*^{1/2} \|u\|_{L^2(I_*; H^{m+2})} < \infty. \quad (18)$$

Equation (1), equivalently

$$u_t = \nu \Delta u - \mathbb{P} \nabla \cdot (u \otimes u),$$

then gives $u_t \in L^1(I_*; H^m)$. Since $u \in L^2(I_*; H^m) \subset L^1(I_*; H^m)$, the Bochner fundamental theorem of calculus gives a unique representative in $C([0, T_*]; H^m)$. $\square$

The full mixed intersection also contains the sharper adjacent coordinates that explain the trace mechanism directly.

Lemma 3.2 (Adjacent Hilbert-scale trace). *Let $s \in \mathbb{R}$. If*

$$v \in L^2(I_*; H^{s+1}), \quad v_t \in L^2(I_*; H^{s-1}),$$

then v has a unique representative in $C([0, T_]; H^s)$. For $0 \leq a \leq b \leq T_*$, this representative satisfies*

$$\left| \|v(b)\|_{H^s}^2 - \|v(a)\|_{H^s}^2 \right| \leq 2 \|v_t\|_{L^2(a,b; H^{s-1})} \|v\|_{L^2(a,b; H^{s+1})}. \quad (19)$$

Proof. The Hilbert scale

$$H^{s+1} \hookrightarrow H^s \hookrightarrow H^{s-1}$$

is a Gelfand triple with pivot H^s ; this is the standard Lions–Magenes trace setting [1]. Time mollification gives, first for smooth functions and then by density,

$$\|v(b)\|_{H^s}^2 - \|v(a)\|_{H^s}^2 = 2 \int_a^b \langle v_t(t), v(t) \rangle_{H^{s-1}, H^{s+1}; H^s} dt.$$

The shifted dual pairing obeys

$$\left| \langle f, g \rangle_{H^{s-1}, H^{s+1}; H^s} \right| \leq \|f\|_{H^{s-1}} \|g\|_{H^{s+1}},$$

so Cauchy–Schwarz proves (19). The same approximation argument gives weak H^s -continuity; weak continuity together with the continuity of the squared norm supplied by the identity gives strong H^s -continuity. Uniqueness follows because two continuous representatives agreeing almost everywhere agree everywhere. $\square$

Applying Lemma 3.2 to every temporal row gives, for all $N, m \in \mathbb{N}_0$,

$$\partial_t^N u \in C([0, T_*]; H^m). \quad (20)$$

Indeed, the two hypotheses are precisely adjacent coordinates of $\mathcal{I}[u_0]$. This is the Lions–Magenes trace mechanism behind the endpoint, whereas Lemma 3.1 shows that the projected row $\mathcal{S}_0(I_*)$ already reaches the same endpoint through the equation.

3.2 The adjacent critical block

Put $\Lambda = (-\Delta)^{1/2}$ and

$$E_s(t) := \frac{1}{2} \|\Lambda^s u(t)\|_2^2, \quad H(t) := E_{1/2}(t). \quad (21)$$

Proposition 3.3 (Critical trace and current). *The adjacent coordinates in $\mathcal{I}[u_0]$ imply*

$$u \in L^2(I_*; H^{5/2}), \quad u_t \in L^2(I_*; H^{1/2}) \implies E_{3/2} \in L^\infty(I_*) \cap L^1(I_*), \quad (22)$$

$$u, u_t \in L^2(I_*; H^1) \implies H \in W^{1,1}(I_*). \quad (23)$$

Moreover the pressure-explicit critical current

$$\mathcal{P}_{1/2} := - \left\langle \Lambda^{1/2} u, \Lambda^{1/2} ((u \cdot \nabla) u + \nabla p) \right\rangle \quad (24)$$

satisfies

$$\mathcal{P}_{1/2} = H' + 2\nu E_{3/2} \in L^1(I_*). \quad (25)$$

Proof. Lemma 3.2 with $s = 3/2$ gives $u \in C([0, T_*]; H^{3/2})$. More precisely, the same time-mollification argument gives, for $0 \leq r \leq t \leq T_*$,

$$E_{3/2}(t) - E_{3/2}(r) = \int_r^t \operatorname{Re}(\Lambda^{1/2} u_t(s), \Lambda^{5/2} u(s))_2 ds. \quad (26)$$

Cauchy–Schwarz therefore gives $E_{3/2} \in W^{1,1}(I_*)$, and in particular $E_{3/2} \in L^\infty(I_*)$. Also

$$\int_0^{T_*} E_{3/2}(t) dt \leq \frac{1}{2} \|u\|_{L^2(I_*; H^{5/2})}^2 < \infty,$$

which proves (22).

The second pair means $u \in H^1(I_*; H^1)$. The Hilbert-space chain rule therefore yields

$$H'(t) = \left\langle \Lambda^{1/2} u(t), \Lambda^{1/2} u_t(t) \right\rangle \quad \text{for a.e. } t, \quad (27)$$

and

$$\|H'\|_{L^1(I_*)} \leq \|u\|_{L^2(I_*; H^1)} \|u_t\|_{L^2(I_*; H^1)} < \infty.$$

Thus $H \in W^{1,1}(I_*)$.

On every compact subinterval of I_* , pair (1) with Λu . The viscous term is $-\nu \|\Lambda^{3/2} u\|_2^2 = -2\nu E_{3/2}$, so

$$H' = -2\nu E_{3/2} + \mathcal{P}_{1/2}.$$

The pressure in (24) is the normalized solution of the Poisson equation in (15); equivalently, its global pairing vanishes because $\nabla \cdot \Lambda u = 0$. The pressure-explicit form is retained because it is the form differentiated in the jet below. Since H' and $E_{3/2}$ are both in $L^1(I_*)$, the classical current extends to the whole terminal interval as the L^1 function in (25). $\square$

No additional critical-current estimate is required after the compatible intersection is present: Proposition 3.3 is a trace consequence of two adjacent rectangles.

3.3 The unique projective endpoint

For each m , let $u_*^{(m)} \in H^m$ denote the endpoint value furnished by Lemma 3.1, equivalently by Lemma 3.2. If $k > m$, then the H^k -continuous and H^m -continuous representatives are the same function almost everywhere on I_* ; after embedding $H^k \hookrightarrow H^m$, continuity makes them identical on $[0, T_*]$. In particular their endpoint values agree in H^m . Therefore the family $\{u_*^{(m)}\}_m$ is compatible and defines one element

$$u_* := u(T_*) \in \bigcap_{m=0}^{\infty} H^m(\mathbb{R}^3) = H^\infty(\mathbb{R}^3). \quad (28)$$

No bound uniform in m is asserted or needed. Since divergence is continuous $H^m \rightarrow H^{m-1}$, u_* is divergence-free; since the H^0 -trace is strong, it also has finite kinetic energy.

The same argument applied to $\partial_t^N u$ in (20) gives compatible endpoint traces

$$V_N^* := \lim_{t \uparrow T_*} \partial_t^N u(t) \in H^\infty, \quad N \in \mathbb{N}_0. \quad (29)$$

3.4 The pressure-explicit endpoint jet

For a multi-index α , set

$$J_{N,\alpha} := \partial_t^N \partial_x^\alpha u, \quad \Pi_{N,\alpha} := \partial_t^N \partial_x^\alpha p.$$

Direct differentiation of (1) gives, on I_* ,

$$\begin{aligned} J_{N+1,\alpha} + \sum_{a=0}^N \sum_{\beta \leq \alpha} \binom{N}{a} \binom{\alpha}{\beta} (J_{a,\beta} \cdot \nabla) J_{N-a,\alpha-\beta} + \nabla \Pi_{N,\alpha} \\ = \nu \Delta J_{N,\alpha}, \quad \nabla \cdot J_{N,\alpha} = 0, \end{aligned} \quad (30)$$

while incompressibility fixes every pressure row through

$$-\Delta \Pi_{N,\alpha} = \partial_i \partial_j \partial_t^N \partial_x^\alpha (u_i u_j). \quad (31)$$

For the pure-time jet write $V_N = \partial_t^N u$ and $Q_N = \partial_t^N p$. Equations (30)–(31) become

$$Q_N = R_i R_j \sum_{a=0}^N \binom{N}{a} V_{a,i} V_{N-a,j}, \quad (32)$$

$$V_{N+1} = \nu \Delta V_N - \sum_{a=0}^N \binom{N}{a} (V_a \cdot \nabla) V_{N-a} - \nabla Q_N. \quad (33)$$

Proposition 3.4 (Endpoint jet reconstruction). *The endpoint velocity u_* uniquely determines a sequence $(V_N^*, Q_N^*)_{N \geq 0} \subset H^\infty \times H^\infty$ by*

$$V_0^* = u_*, \quad (34)$$

$$Q_N^* = R_i R_j \sum_{a=0}^N \binom{N}{a} V_{a,i}^* V_{N-a,j}^*, \quad (35)$$

$$V_{N+1}^* = \nu \Delta V_N^* - \sum_{a=0}^N \binom{N}{a} (V_a^* \cdot \nabla) V_{N-a}^* - \nabla Q_N^*. \quad (36)$$

This recursively constructed sequence equals the left endpoint traces of the actual velocity-pressure jet.

Proof. The Fréchet space H^∞ is closed under spatial differentiation and pointwise multiplication. Riesz transforms preserve each H^m , hence preserve H^∞ . Thus (35)–(36) construct one and only one H^∞ pair at every order.

It remains to identify this formal recursion with the history. Fix m . The traces (29) converge in arbitrarily high Sobolev spaces. Sobolev multiplication, spatial differentiation, and the Riesz transforms are continuous between the corresponding spaces. We may therefore pass $t \uparrow T_*$ in (32)–(33). Beginning with $V_0^* = u_*$, induction in N gives exactly (35)–(36). Since m was arbitrary, the identities hold in H^∞ . Spatial differentiation then identifies the full multi-index jet in (30). $\square$

3.5 Classical restart and contradiction of finite maximality

We record the standard local theorem in precisely the form used here [2].

Lemma 3.5 (Local strong solution and uniqueness). *Let $s > 5/2$, let $t_0 \in \mathbb{R}$, and let $a \in H^s(\mathbb{R}^3)$ be divergence-free. There is a $\delta = \delta(\nu, s, \|a\|_{H^s}) > 0$ and a unique solution of the Navier–Stokes equation on $[t_0, t_0 + \delta]$, with $w(t_0) = a$, such that*

$$w \in C([t_0, t_0 + \delta]; H^s) \cap L^2(t_0, t_0 + \delta; H^{s+1}).$$

If $a \in H^\infty$, every Sobolev order persists on a common possibly smaller interval and (w, q) is smooth up to $t = t_0$. Its temporal jet at t_0 is the pressure-normalized recursion (35)–(36) with $V_0^ = a$.*

Proof. The Friedrichs approximations satisfy the standard commutator estimate

$$\frac{1}{2} \frac{d}{dt} \|w\|_{H^s}^2 + \nu \|\nabla w\|_{H^s}^2 \leq C_s \|\nabla w\|_{L^\infty} \|w\|_{H^s}^2.$$

Since $s > 5/2$, Sobolev embedding controls $\|\nabla w\|_{L^\infty}$ by $\|w\|_{H^s}$. A uniform short-time bound, compactness of the regularized sequence, and the equation give existence with the displayed regularity. Applying the corresponding estimate to the difference of two solutions and using Gronwall proves

uniqueness. Repeating the estimate after differentiating the equation gives persistence of each H^m norm; a diagonal choice of the same base lifespan works because the local time is fixed by the H^s bound, while the higher estimates are linear once that base solution is known. Finally, differentiating the equation at the initial time and using the normalized pressure Poisson equation yields the stated jet recursion. $\square$

Apply Lemma 3.5 at time T_* with datum $a = u_*$. It gives a smooth solution (w, q) on $[T_*, T_* + \delta]$. Define

$$(\tilde{u}, \tilde{p})(t) := \begin{cases} (u, p)(t), & 0 \leq t < T_*, \\ (w, q)(t), & T_* \leq t \leq T_* + \delta. \end{cases}$$

The velocity traces agree at T_* by construction. Therefore the piecewise function has no distributional time-boundary defect and solves (1) across T_* . Proposition 3.4 and the last assertion of Lemma 3.5 show that every left and right velocity-pressure derivative agrees at T_* . Hence $(\tilde{u}, \tilde{p})$ is classical, indeed smooth, on an interval strictly larger than $[0, T_*)$.

Equivalently, one may start the local solution at a time $t_0 < T_*$ sufficiently close to T_* . The continuous H^s trace gives a uniform local lifespan, and strong-solution uniqueness identifies that solution with u on the overlap. Both descriptions restart the same datum-generated history.

This contradicts the definition of T_* as the maximal classical lifespan. Thus the assumption $T_* < \infty$ is impossible, and

$$\boxed{T_* = \infty.}$$

Because the argument applies at every finite time and all velocity and pressure rows lie in H^∞ , Sobolev embedding yields

$$\boxed{u, p \in C^\infty([0, \infty) \times \mathbb{R}^3).}$$

This completes every implication in (2) and proves Theorem 1.1.

References

- [1] J.-L. Lions and E. Magenes, *Non-Homogeneous Boundary Value Problems and Applications, Vol. I*, Springer, 1972.
- [2] R. Temam, *Navier–Stokes Equations: Theory and Numerical Analysis*, AMS Chelsea, 2001.